

SARA CHOPRA

Hamburg , Germany || Science Communication & Physics

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Physicist and science communicator with a background in theoretical particle physics and research-driven public outreach. Experienced in translating complex scientific concepts into accessible visual and digital formats, including interactive web explainers and multimedia content. Currently developing web-based communication projects for the Cluster of Excellence “Advanced Imaging of Matter” (CUI) while expanding skills in machine learning and AI-assisted workflow development. Seeking roles that combine scientific rigour with communication, digital & design strategy, and AI-assisted workflows.

Professional Experience

Student Assistant (PR & Communication Department)

03/2024 - 03/2026

Cluster of Excellence for Advanced Imaging (CUI), Universität Hamburg

- Managing social media content and strategy for varied audiences across [Instagram](#), [LinkedIn](#), and web, including research interview and lab tour video production (ideation to production to editing) and supported outreach events on-site, eg. [Science City Day](#).
- Designing outreach assets for events and digital platforms (posters, graphics, reels/videos, print materials).[\(See samples here\)](#)
- Provide structured office + analytical support: maintain internal datasets for [projects](#) along with performed event feedback evaluation.
- Gave a lecture to high school students on “[Particle and Accelerator Physics](#)” for [Lights and Schools](#).
- Recently, contributed to a [CUI news piece](#) for a latest research publication. Currently developing a CUI web infographic explaining the different research areas.

Independent Project: [CMS Detector Web Infographic](#)
Scrollytelling webpage explaining detector components and physics. Built with HTML/CSS/JS; used AI assistance to speed up implementation; content inspired by CERN/CMS public resources. (10 day build)

Communication & Media

- Science communication for specialist & non-specialist audiences.
- Social media content + strategy
- Short-form video production/editing + interview coordination

Design & Digital

- Canva, Adobe Creative Suite
- Blender (for 3D scientific visualisation)
- MS Office
- Web basics: HTML/CSS

Languages

English: Fluent (Native level),
German: Level (B2.1) Intermediate,
Hindi: Native

Conferences & Workshops

- KMBP School on QFT, Higher Topological Algebra, and Gauge Symmetry (2024)
- DPG (Deutsche Physikalische Gesellschaft) Conference, 2024
- NIUS Camp (Homi Bhabha Centre for Fundamental Research)
- Science Communication Workshop, Humboldt University (Dr. Alison Pinto, 2024)

Scholarships/Awards

- Merit Scholarship for International Students: 2024 - 2025 (Universität Hamburg)
- 50% Academic Fee Scholarship in Bachelor's: 2019 & 2021 (Amity Univeristy)
- Goethe Institute: Scholarship Program (Sri Lanka): June 2017

Education

10/2022 - 04/2026

Universität Hamburg,
Germany

Masters in Physics

Accelerator and Particle
Physics (specialisation)

Master Thesis Project :

‘Sum Rules in Extended Higgs
Model and their Phenomenological
Applications

Master’s Result: (German System)
(Grade Range 1-6 (1- highest grade))

Final Result : 1.29

2019 - 2022

AIAS, Amity University,
Noida, India

B.Sc. (Honors) Physics

Bachelor’s Result: (Indian Grading System)
(10 is the highest grade)

1st Year: 9.52/10
2nd Year: 9.38/10
3rd Year: 9.30/10

2005 - 2019

Springdale’s School, Pusa
Road, New Delhi, India

Higher Secondary Schooling

High School Result: (Indian Grading System)
(100% is the highest grade)

12th Standard: 91.5%
Science Stream with Economics & German
10th Standard: CGPA: 10/10

Technical & Analytical Skills

- Specialised in Theoretical and Phenomenological particle Physics & Computational Analysis.
- Programming & Data Analysis
- Scientific Writing & Documentation

Technical Software Used:

Mathematica (FeynCalc, FeynArts, SARAH), Python, SQL, LaTeX.

Volunteering & Extracurricular Activities

- Volunteer at eVidyaloka (2022): Taught and created educational content for students in rural areas.
- Each One Teach Many Program (2014-2019): Provided education to eight underprivileged children.

Hobbies and Interests

- **Art & Design:** Sketching, acrylic & oil painting
- **Media & Storytelling:** Video editing, science interviews
- **Music, Cooking, Travel:** Engaging with cultures and creative expression
- **Psychology and Neuroscience**

Research Experience

Master Thesis

07/2023 - 04/2026

Sum Rules in Extended Higgs Models and Their Phenomenological Applications, Institute of Theoretical Physics II , Universität Hamburg

Supervisors: Dr. Gudrid Moortgart Pick and Dr. Sven Heinemeyer

- Conducted theoretical and phenomenological analysis of extended Higgs sector models (2HDM, Georgi Machacek, Septet).
- Applied Mathematica and Python packages to evaluate phenomenological constraints and to conduct comparative analysis.
- Compared theoretical predictions and consistency of BSM models to collider observables at the LHC and computing allowed and excluded cross sections using HiggsTools.

Research Internship: Humboldt University, Berlin (2024)

10/2024 - 01/2025

Under Dr. Olaf Hohm (RTG 2575: Rethinking Quantum Field Theory)

- Studied homotopy transfer and BV algebras for gauge-invariant formulations of the path integral; extended the BV formalism to 4-term complexes with gauge symmetries.
- Investigated Chevalley–Eilenberg algebras and their role in deriving the Faddeev–Popov determinant; participated in weekly seminars on higher algebra and QFT.

Volunteering and Leadership Experience

Vice President and Member:

2019 - 2022 (Member 3 years, VP: 2020 - 2021)

KugelBlitz Physics Club, Amity University

- Led the transition from offline to online events, successfully implementing activities like research discussions, virtual treasure hunt, and board games.
- Established the club's organizational structure and social media presence, implementing a strategy that increased event audience from 5 to 50-75 participants.
- Acquired sponsorships and expanded active membership from 10 to 25, strengthening community engagement.
- Organised 20+ events and led brainstorming sessions, fostering a culture of innovation and participation.

Content & Design Team Member

04/2020 - 09/2020 (6 months)

Volunteer at NGO: GirlUp Empower

- Created over 10 articles on topics like women in science, contemporary beauty standards, and refining research and writing skills.
- Designed around 5-10 social media posts weekly, ensuring visually appealing and informative content for a broader audience.
- Developed an understanding of aesthetics, creative storytelling, and audience engagement through marketing-focused design.